Introduction to Credit Risk and Applications in Python

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Lecture Outline

Introduction

- Most of the time, portfolios appear stable.
- Observed default rates are low.
- Losses cluster around small expected values.

Why Study Credit Risk?

If Defaults Are Rare, Where Does Systemic Risk Come From?

- **Dependence —** Defaults do not occur independently.
- **Factor structure —** Common economic drivers create clustering.
- **Model assumptions —** Distributional choices shape outcomes.
- **Tail amplification —** Small structural effects become large in extremes.

Dependence and Tail Amplification: A Simulation

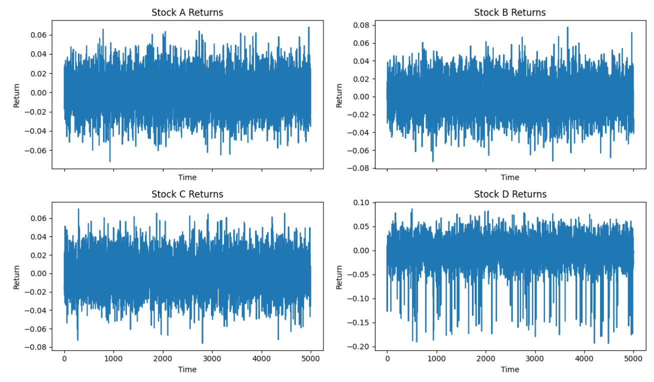


Figure 1: Simulated Multivariate Returns with Correlation and Tail Shock

Uncomfortable Facts

- **Fact 1 — Credit risk is usually invisible in calm periods.**
- **Fact 2 — Correlation drives extreme portfolio losses.**
- **Fact 3 — Model assumptions determine measured capital.**

From Stability to Systemic Risk

- Risk is not about average losses.
- Risk is about rare joint events.
- Banks do not fail because a single borrower defaults.
- Banks fail when many exposures move together.
- Regulatory capital is determined by assumptions about dependence and tail behavior.

What This Course Is About (Modeling View)

- **1. Modeling individual default.** Describe whether a firm defaults as a random outcome driven by its underlying financial condition.
- **2. From individual firms to a portfolio.** Aggregate many firms to understand how individual defaults translate into total portfolio losses.
- **3. Comovement and systemic risk.** Capture how firms are jointly affected by common economic factors, leading to correlated defaults.
- **4. Risk management and capital.** Quantify potential losses and determine how much capital is needed to absorb extreme outcomes.

Credit Risk Measurement

Python

Python in This Course

Python for Credit Risk Modelling

In this course, Python is used as a **research environment**, not as a programming course.

Focus:

- implementing quantitative models
- running simulations
- analysing risk distributions
- producing reproducible results

Goal: Use programming as a tool for financial modelling.

Course Philosophy

Modern coding tools and AI can assist with writing code.

However, successful quantitative work still requires:

- understanding model structure
- debugging incorrect implementations
- designing simulations
- interpreting statistical outputs

Key principle:

Programming is a **tool for modelling and analysis**, not an end in itself.

Development Environment Used in This Course

In this course we standardize the working environment to simplify collaboration.

Two main tools will be used:

- **PyCharm** as the primary Python development environment
- **GitHub** as the collaboration and version control platform

Python: Setup

Installing Python

Python must be installed before setting up the development environment.

Step 1: Download Python

- Visit: `https://www.python.org`
- Download the latest stable version of Python
- Recommended: Python 3.x

Step 2: Install Python

- Run the installer
- Select **Add Python to PATH**
- Complete the installation

Verification

Installing PyCharm

PyCharm will be used as the main development environment.

Step 1: Download PyCharm

- Visit: `https://www.jetbrains.com/pycharm`
- Download **PyCharm Community Edition**

Step 2: Install the IDE

Step 3: Create a Project

- Open PyCharm
- Click **New Project**
- Select the installed Python interpreter

Git: Setup

Installing Git

Git is required to interact with GitHub repositories.

Step 1: Download Git

- Visit: `https://git-scm.com`
- Download the installer for your operating system

Step 2: Install Git

- Run the installer
- Accept default configuration options

Verification

```
git --version
```

Creating a GitHub Account

GitHub will be used for collaboration.

Step 1: Create an Account

- Visit: `https://github.com`
- Click **Sign Up**
- Create a username and password

Step 2: Configure Git

After installing Git, configure your identity:

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

Workflow

Structured Projects

Typical structure:

```
credit-risk-project/

 data/
 src/
    simulation.py
    models.py
    utils.py
 notebooks/
 results/
 requirements.txt
 README.md
```

Principle: Separate data, code, analysis, and results.

Virtual Environments

Python environments isolate project dependencies.

Benefits:

- consistent package versions
- fewer compatibility issues
- easier replication of results

Common tools:

- `venv`
- `conda`

Dependencies are stored in

`requirements.txt`

Why Version Control?

Without version control:

- files get overwritten
- changes are difficult to track
- collaboration becomes unreliable

Version control systems record:

- who changed code
- what changed
- when it changed

Standard tool: **Git**

Basic Git Workflow

Typical development workflow:

1. Clone the repository
2. Create a new branch
3. Implement a feature
4. Commit changes
5. Submit a pull request

Benefits:

- changes are documented
- work can be reviewed
- errors can be reverted

Collaboration on GitHub

GitHub provides infrastructure for team development.

Main collaboration tools:

- pull requests
- issue tracking
- code review

Typical workflow:

```
feature branch → pull request → review → merge
```

Team Assignment Structure

Students work in **teams of two to three**.

Project setup:

- Each student creates an individual branch in the repository: `https://github.com/Zazazzzzz/Credit-Risk-Student-Workspace` (*send an email to be added as a collaborator*)
- Each team creates a shared project file and starts developing a script collaboratively
- Submit an initial test contribution (e.g., small script or file) to understand the collaboration workflow
- Use this setup throughout the course to jointly develop and submit the final assignment

Course Development Workflow

Initial Setup:

`Install Python` → `Install PyCharm (IDE)` → `Install Git` → `Create GitHub Account`

After the setup:

`Create Project` → `Write Code` → `Commit` → `Push` → `Pull Request`

Objective: Develop models collaboratively while maintaining reproducible and well-structured research code.

Python: Programming

Core Python Libraries

Key tools used in this course:

NumPy

- vectorized computations
- random number generation

Pandas

- portfolio data handling

SciPy

- probability distributions

Matplotlib / Seaborn

- visualization of results

Programming Focus

The course emphasizes:

- Monte Carlo simulation design
- modular functions
- clear model implementation
- efficient numerical computation

The goal is to translate **financial models into executable code**.

Basic Statistics

Why statistics in credit risk?

- Credit risk models produce **random losses**.
- Risk measures (VaR, ES) are **features of distributions**.
- Dependence drives **clustering** and tail risk.

Statistics → Probability → Simulation

Random Variables: The Core Object

A random variable assigns a number to a random outcome.

Credit risk examples

- Y_i: default indicator
- K: number of defaults
- L: portfolio loss
- X_i: latent (asset/creditworthiness) variable

Default indicator

$$Y_i = \begin{cases} 1 & \text{default} \\ 0 & \text{no default} \end{cases}$$

$$P(Y_i = 1) = p_i$$

Random Variable: Discrete vs. Continuous

Discrete

- takes values in a countable set
- examples: default indicator Y_i
- probabilities: $P(Y = 1) = p, \quad P(Y = 0) = 1 - p$

Continuous

- takes values on a continuous range (intervals)
- examples: returns, latent variable X_i
- $P(X = x) = 0$ for any single value x
- described by densities, e.g. $X \sim \mathcal{N}(\mu, \sigma^2)$

Key Distributions I: Bernoulli, Binomial, Poisson

Bernoulli Distribution (one obligor)	**Binomial Distribution** (number of defaults)	**Poisson Distribution** (rare events)
$Y_i \sim \text{Bernoulli}(p_i)$	$K = \sum_{i=1}^{m} Y_i$	$K \sim \text{Poisson}(\lambda)$
$Y_i \in \{0, 1\}$	$K \sim \text{Binomial}(m, p)$	$\lambda = mp$

Independent defaults $\Rightarrow$ Binomial

Large m, small p $\Rightarrow$ Poisson approximation

Key Distributions I: Probability Mass Functions

Probability Mass Functions

- **Bernoulli:**

$$P(Y_i = y) = p_i^y (1 - p_i)^{1-y}, \quad y \in \{0, 1\}$$

- **Binomial:**

$$P(K = k) = \binom{m}{k} p^k (1 - p)^{m-k}, \quad k = 0, 1, \ldots, m$$

- **Poisson:**

$$P(K = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

- PMF assigns probability to each possible outcome.

Key Distributions II: Normal Distribution

Why normal matters:

- central building block for factor models and Gaussian copula
- convenient for thresholds via $\Phi^{-1}(\cdot)$

$$X \sim \mathcal{N}(\mu, \sigma^2), \qquad Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Probability Density Function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Key Distributions III: Beta and Gamma

Beta Distribution

- Defined on the interval $[0, 1]$
- Flexible shapes for modelling probabilities
- Useful when the probability itself is uncertain

$$p \sim \mathrm{Beta}(\alpha, \beta)$$

Application: modelling uncertainty in PD.

Gamma Distribution

- Positive-valued distribution
- Flexible skewness and tail behaviour
- Often used for random intensities

$$\lambda \sim \mathrm{Gamma}(k, \theta)$$

Application: default intensity models.

These distributions often appear as mixing distributions in credit risk models.

Mixtures and Overdispersion I

In practice, default probabilities are rarely known exactly.

Idea: model the PD itself as random.

$$p \sim \mathrm{Beta}(\alpha, \beta)$$

Conditional on p, defaults follow

$$K \mid p \sim \mathrm{Binomial}(n, p)$$

This leads to a **Beta–Binomial mixture**:

$$K \sim \text{Beta-Binomial}$$

Implication

- higher variance than a binomial model; more frequent clusters of defaults
- captures **overdispersion in credit portfolios**

Mixtures and Overdispersion II

In some models, defaults are viewed as **arrival events over time**.

Poisson model

$$N \mid \lambda \sim \text{Poisson}(\lambda)$$

where λ is the default arrival intensity.

Allow the intensity to be random

$$\lambda \sim \text{Gamma}(k, \theta)$$

Resulting mixture distribution:

$$N \sim \text{Negative Binomial}$$

Implication

- higher variance than Poisson; more clustered default events
- captures **overdispersion in default arrivals**

Mixtures and Overdispersion III

In practice, defaults are driven by an underlying continuous variable.

Idea: link a continuous distribution to a discrete outcome.

$$X_i \sim \mathcal{N}(0,1)$$

Default occurs if:

$$Y_i = \mathbf{1}(X_i \leq d_i)$$

Where

- X_i represents a latent variable (e.g. asset return or firm condition)
- Y_i is a binary outcome derived from X_i

Implication

- introduces **randomness** through an underlying continuous driver; provides a foundation for **dependence and clustering** via the latent variable

Idea: The expected value describes the long-run average outcome.

$$\mathbb{E}[X]$$

Default Indicator

$$Y_i \sim \text{Bernoulli}(p_i)$$
$$\mathbb{E}[Y_i] = p_i$$

Portfolio Expected Loss

$$EL = \sum_{i=1}^{m} w_i \, LGD_i \, p_i$$

Expected loss is an average outcome, not a stress scenario.

Variance and Standard Deviation: Uncertainty Around the Mean

Idea: variance measures dispersion; standard deviation is dispersion in original units.

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right], \qquad \sigma(X) = \sqrt{\mathrm{Var}(X)}$$

Bernoulli example (default indicator):

$$\mathrm{Var}(Y_i) = p_i(1 - p_i)$$

Takeaway: low PD does not imply low uncertainty in portfolio outcomes.

Covariance and Correlation: Dependence

Covariance: joint movement (scale-dependent).

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big]$$

Correlation: standardized dependence (-1 to 1).

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$$

Credit risk message: dependence $\Rightarrow$ clustering $\Rightarrow$ heavier loss tails.

Cumulative Distribution Function (CDF)

Definition:

$$F_L(x) = P(L \leq x), \quad x \in \mathbb{R}$$

Properties:

- $0 \leq F_L(x) \leq 1$
- $F_L(x)$ is non-decreasing in x
- $\lim_{x \to -\infty} F_L(x) = 0, \quad \lim_{x \to \infty} F_L(x) = 1$

Interpretation:

- $F_L(x)$ gives the probability that the loss does not exceed x
- fully characterizes the distribution of L

Quantiles: From Distributions to Risk Measures

Quantile definition:

$$q_\alpha(L) = \inf\{x \in \mathbb{R} : F_L(x) \geq \alpha\}, \quad \alpha \in (0,1)$$

Interpretation:

- $q_\alpha(L)$ is the smallest loss level such that the cumulative probability reaches α
- quantiles characterize tail behavior without symmetry assumptions

First Portfolio Simulation

Before introducing theoretical models, we begin with a simple simulation.

Example portfolio:

- A bank lends **€100,000 each to 10 firms**.
- All firms promise to repay the full amount in one year (no interest, single payment).
- Each firm may default:
  - Firm 1: 1% chance of default
  - Firm 2: 2%
  - …
  - Firm 10: 10%
- If a firm defaults, it only repays **€50,000**
  $\Rightarrow$ the bank loses €50,000

What Do We Want to Understand?

On average:

- How much loss should the bank expect?

In reality:

- Losses are uncertain
- If we repeat this situation **1,000 times (simulation)**, what does the **distribution of total losses** look like?

Simulation Workflow

Simulation steps:

1. simulate default indicators
2. compute losses for each obligor
3. aggregate portfolio losses
4. repeat for many scenarios
5. analyse the loss distribution

This introduces key concepts used later:

- random variables
- probability distributions
- tail risk

Motivation for Credit Risk Models

The simulation illustrates an important question:

How do we model dependence between defaults?

This motivates the models studied later:

- threshold models
- factor models
- copulas
- portfolio loss distributions

Credit Risk

Introduction

What is Credit Risk?

Credit risk is the risk of a loss arising from the failure of a counterparty to honor its contractual obligations. This subsumes both default risk (the risk of losses due to the default of a borrower or a trading partner) and downgrade risk (the risk of losses caused by a deterioration in the credit quality of a counterparty that translates into a downgrading in some rating system).

Obligor: A counterparty with a financial obligation to another party. For example, a debtor who owes money.

Default: The failure to fulfill a financial obligation, such as repaying a loan or paying interest/coupons on a bond.

Basic Concepts

Some fundamental definitions in this context:

- **Probability of Default (p):** the likelihood that a counterparty will fail to meet their debt obligations over a specific time horizon.
- **Exposure at Default (EAD):** measures the total outstanding that is at risk if the borrower defaults.
- **Loss Given Default (LGD):** the percentage of the total exposure that is expected to be lost in the event of default.
- **Expected Loss (EL):** represents the average or anticipated loss that a financial institution expects to incur over a specified period due to defaults and credit impairments.
- **Unexpected Loss (UL):** the loss that exceeds the average expected loss.

Expected vs. Unexpected Loss

Expected Loss (EL)

$$EL = \sum_{i=1}^{m} w_i \, LGD_i \, p_i$$

- Average loss expected over the time horizon
- Driven by:
  - Probability of default p_i
  - Loss given default LGD_i
  - Exposure weight w_i
- Typically covered by pricing, provisions, and reserves

Unexpected Loss (UL)

- Deviation of actual losses from the expected level
- Arises from:
  - Default uncertainty
  - Correlation between obligors
  - Systematic economic shocks
- Managed using **economic and regulatory capital**

The Importance of Dependence Structure

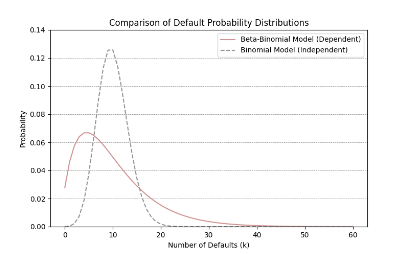


Figure 2: Comparison of the Distribution of Default Numbers of Different Dependence Structures

This plot is a comparison of the loss distribution of a homogeneous portfolio of 1000 loans with a default probability of $p_1 = \cdots = p_{1000} = 1\%$ assuming (i) independent defaults and (ii) a default correlation of 0.5%.

The Importance of Dependence Structure

There are two primary sources of dependence between defaults:

- **Common Factor Dependence:** Defaults may be influenced by shared macroeconomic factors, such as interest rates, GDP growth, or inflation, which impact all obligors simultaneously.
- **Contagion Effects:** The default of one company (e.g., Company A) can directly affect the default probability of another company (e.g., Company B) due to interconnected business relationships, financial linkages, or supply chain dependencies, a phenomenon known as contagion.

OUTCOME: When defaults are correlated, extreme losses become more likely, leading to higher tail risk in the portfolio's loss distribution. In reality, it can cause economic downturns, liquidity crises, or sector-specific shocks.

Default Modeling

Default as a Random Event

Credit risk begins with a simple question:

Does the obligor default over the horizon?

$$Y_i = \begin{cases} 1, & \text{default} \\ 0, & \text{no default} \end{cases}$$

Implications

- Each obligor is reduced to a binary outcome.
- The randomness comes from not knowing in advance which firms will default.
- Portfolio credit risk is built by combining many such indicators.

Threshold Representation of Default

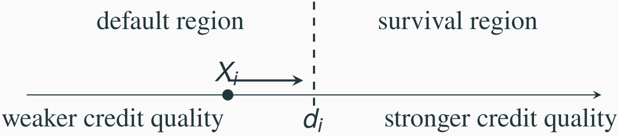
Default rule

$$Y_i = \begin{cases} 1, & \text{if } X_i \le d_i, \\ 0, & \text{if } X_i > d_i \end{cases}$$

- X_i: latent variable measuring the obligor's credit quality
- d_i: default threshold
- Default occurs when credit quality falls below the threshold

Interpretation

-**High** X_i: firm remains healthy
- **Low** X_i: firm is under distress
- **Threshold crossing**: default event



Structural Model

Merton's Model

Merton's model (1974) serves as the prototype of all firm value models. Consider a firm with a stochastic asset value V_t, financing itself through equity (i.e., by issuing shares) and debt.

- The firm's debt consists of a single zero-coupon bond with a face (or nominal) value B and maturity T.
- Let S_t and B_t denote the values of equity and debt, respectively, at time $t \leq T$.
- The firm's total value satisfies the relationship:

$$V_t = S_t + B_t, \quad 0 \leq t \leq T$$

- Default occurs if the firm fails to meet its payment obligations to debt holders, which can only happen at time T.

Merton's Model

Possible Cases at T:

a. $V_T > B$:
   - The debtholders receive B.
   - The shareholders receive the residual value $S_T = V_T - B$, and there is no default.

b. $V_T \leq B$:
   - The firm cannot meet its financial obligations.
   - Shareholders hand over control to the bondholders, who liquidate the firm.
   - In this case, $B_T = V_T$ and $S_T = 0$.

In summary, we obtain:

$$S_T = (V_T - B)^+$$

$$B_T = \min(V_T, B) = B - (B - V_T)^+$$

The Asset Value Process:

- It is assumed that the asset value V_t follows a geometric Brownian process of the form:

$$dV_t = \mu_V V_t \, dt + \sigma_V V_t \, dW_t \tag{1}$$

  for constants $\mu_V \in \mathbb{R}$, $\sigma_V > 0$, and a standard Brownian motion $(W_t)_{t \geq 0}$.
- The solution at time T of the stochastic differential equation (1) is given by:

$$\ln V_T = \ln V_0 + \left(\mu_V - \frac{1}{2}\sigma_V^2\right) T + \sigma_V W_T, \tag{2}$$

$$V_T = V_0 \exp\left(\left(\mu_V - \frac{1}{2}\sigma_V^2\right) T + \sigma_V W_T\right) \tag{3}$$

Default Probability:

From equation (2), we know $\ln V_T \sim N\left(\ln V_0 + \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T, \sigma_V^2 T\right)$.

The default probability is therefore:

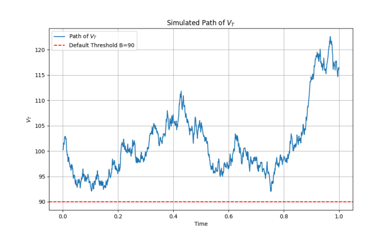
$$\begin{aligned} P(V_T \leq B) &= P(\ln V_T \leq \ln B) \\ &= \Phi\left(\frac{\ln B - \ln V_0 - \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}\right). \end{aligned} \tag{4}$$

The Merton model is equivalent to the threshold model (X, d) with

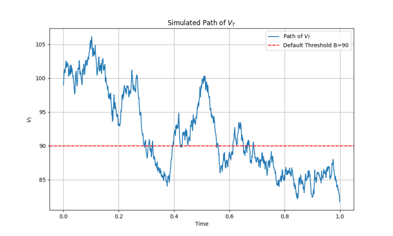
$$X := \frac{\ln V_T - \ln V_0 - \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}},$$

$$d := \frac{\ln B - \ln V_0 - \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}.$$

Merton's Model



(a) Path without default.



(b) Path with default

Figure 3: Comparison of default paths for V_T.

Task 1: Simulate default path under Merton's model

To simulate Figure 3a.

Parameters: `V_0 = 100, mu_V = 0.05, sigma_V = 0.2, T = 1.0, B = 90, N = 1000, dt = T / N, np.random.seed(42).`

Portfolio

Portfolio Setting

- We consider a portfolio of m obligors and fix a time horizon T.
- The random vector $\mathbf{Y} = (Y_1, \ldots, Y_m)'$ is a vector of default indicators for the portfolio.
- The marginal default probabilities are denoted by $p_i = P(Y_i = 1)$, where $i = 1, \ldots, m$.
- The joint probability is $p(\mathbf{y}) = P(Y_1 = y_1, \ldots, Y_m = y_m), \quad \mathbf{y} \in \{0, 1\}^m$

Task 2: Simulate default outcome in a portfolio

Simulate 1000 trials of a portfolio consisting of 100 obligors, each with a 1% probability of default, assuming the default events are independent. Then, plot the density of the number of defaults observed in the simulations.

Parameters:

`N = 1000, P(Y_i) = 1%, m = 100, np.random.seed(42).`

Portfolio Setting

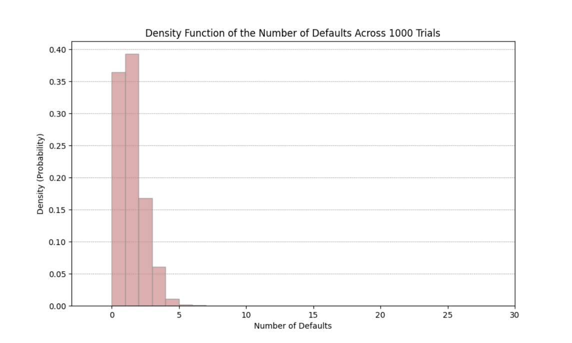


Figure 4: Simulation of Default Outcomes of a Portfolio With No Interdependence: Homogeneous obligors

Dependency Structure in a Portfolio

In the same setting, we derive the **default correlation.** Noting that

$$\mathrm{var}(Y_i) = \mathbb{E}(Y_i^2) - p_i^2 = \mathbb{E}(Y_i) - p_i^2 = p_i - p_i^2,$$

we obtain, for firms i and j with $i \neq j$, the formula

$$\rho(Y_i, Y_j) = \frac{\mathbb{E}(Y_i Y_j) - p_i p_j}{\sqrt{(p_i - p_i^2)(p_j - p_j^2)}}. \tag{5}$$

Task 3: The dependency structure of the obligors

Y_i for $i = 1,\ 2$ is the default indicator during a time period for two companies. Now, assume we have the following joint probabilities:

$$P(Y_1 = 0, Y_2 = 0) = 0.85 \quad P(Y_1 = 1, Y_2 = 1) = 0.05$$
$$P(Y_1 = 1, Y_2 = 0) = 0.07 \quad P(Y_1 = 0, Y_2 = 1) = 0.03$$

- Compute the marginal probabilities of default for Y_1 and Y_2. And the default correlation.
- Compute the joint probability of default, that is $P(X_1 \leq d_1,\ X_2 \leq d_2)$ (threshold model), that is the $C(p_1, p_2)$.

Marginal and Joint Distribution

The given probabilities describe the *joint distribution* of (Y_1, Y_2). That is, they describe the probabilities of all possible combinations of default and non-default for the two obligors.

The *marginal default probability* of one obligor is obtained by summing over all outcomes of the other obligor.

For Y_1, we have

$$p_1 = P(Y_1 = 1) = P(Y_1 = 1, Y_2 = 0) + P(Y_1 = 1, Y_2 = 1).$$

Therefore,

$$p_1 = 0.07 + 0.05 = 0.12.$$

For Y_2, we have

$$p_2 = P(Y_2 = 1) = P(Y_1 = 0, Y_2 = 1) + P(Y_1 = 1, Y_2 = 1).$$

The *default correlation* measures the dependence between the default events Y_1 and Y_2. It is defined as according to (5):

$$\frac{P(Y_1 = 1, Y_2 = 1) - P(Y_1 = 1)P(Y_2 = 1)}{\sqrt{P(Y_1 = 1)(1 - P(Y_1 = 1))P(Y_2 = 1)(1 - P(Y_2 = 1))}}.$$

Substituting the values gives

$$\rho(Y_1, Y_2) = \frac{0.05 - (0.12)(0.08)}{\sqrt{0.12(1 - 0.12) \cdot 0.08(1 - 0.08)}}.$$

Hence,

$$\rho(Y_1, Y_2) = \frac{0.0404}{0.0883} \approx 0.457.$$

Joint Default Probability

The *joint probability of default* is

$$P(Y_1 = 1, Y_2 = 1) = 0.05.$$

Using the marginal default probabilities

$$p_1 = P(X_1 \leq d_1), \qquad p_2 = P(X_2 \leq d_2),$$

the copula representation gives

$$P(X_1 \leq d_1,\ X_2 \leq d_2) = C(p_1, p_2).$$

Thus,

$$C(p_1, p_2) = P(Y_1 = 1, Y_2 = 1) = 0.05.$$

Default correlation and asset correlation

- The probability of default for an obligor is given by:

$$P(Y_i = 1) = P(X_i \le d_i) = F_{X_i}(d_i).$$

- Default correlation: For given default probabilities, the correlation between defaults $\rho(Y_i, Y_j)$ is determined by $\mathbb{E}(Y_i Y_j)$ according to (5). In a threshold model, $\mathbb{E}(Y_i Y_j) = P(X_i \le d_i, X_j \le d_j)$, so the default correlation depends on the joint distribution function of X_i and X_j.
- Asset correlation: Since the critical variables are often interpreted in terms of asset values, the correlation of the critical variables X_i and X_j is often referred as asset correlation.

Why Asset Correlation Is Not Default Correlation

Even if X_i and X_j are strongly correlated, defaults only occur in the lower tail:

$$X_i \le d_i, \qquad X_j \le d_j.$$

Therefore, default correlation depends not only on $\rho(X_i, X_j)$, but also on the default thresholds d_i, d_j, or equivalently the default probabilities p_i, p_j.

$$P(Y_i = 1, Y_j = 1) = P(X_i \le d_i,\ X_j \le d_j).$$

So the same asset correlation can lead to different default correlations if the marginal default probabilities are different.

$$\rho(X_i, X_j) \neq \rho(Y_i, Y_j)$$

Dependency Structure in a Portfolio

Asset Correlation: Measures the correlation between the latent variables X_i. It is a parameter of the factor model and directly controls the dependence structure of the X_i.

Default Correlation: Measures the correlation between default events Y_i. It is influenced by the asset correlation and the marginal probability of default p_i.

Dependency Structure in a Portfolio

We consider a special case: **an exchangeable default model**.

- It is common to group obligors together to form homogeneous groups. This corresponds to the mathematical concept of exchangeability.
- Note that this permits a simple notation for default probabilities:

$$\pi_k := P(Y_{i_1} = 1, \ldots, Y_{i_k} = 1), \quad \{i_1, \ldots, i_k\} \subset \{1, \ldots, m\}, \tag{6}$$

$$\pi := \pi_1 = P(Y_i = 1), \quad i \in \{1, \ldots, m\}. \tag{7}$$

- In the exchangeable case, the default correlation is given by

$$\rho_Y := \rho(Y_i, Y_j) = \frac{\pi_2 - \pi^2}{\pi - \pi^2}, \quad i \neq j.$$

Factor Models

Why Factor Models?

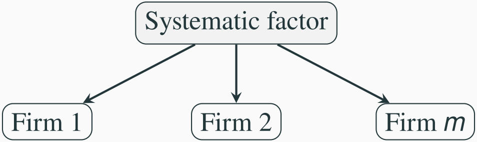
Economic reality

- Firms are exposed to the same macroeconomic environment
- Economic shocks affect many companies simultaneously
- Defaults therefore tend to **cluster**

Defaults are not independent events.

Goal of factor models

- Introduce **systematic risk factors**
- Allow many obligors to respond to the same shock
- Generate **correlated defaults** in a tractable way



One systematic shock → many obligors

Latent Variable Factor Models

Latent Variable Approach

For an obligor i, the **critical variable X_i** is modeled as:

$$X_i = \rho_i \tilde{F}_i + \sqrt{1 - \rho_i^2}\epsilon_i, \tag{8}$$

where:

- X_i: Critical variable for obligor i. A lower value of X_i indicates higher credit risk.

- $\tilde{F}_i$: Common factor (systematic risk), typically assumed to follow a normal distribution.

- ρ_i: Sensitivity of obligor i to the common factor (factor loading).

- ϵ_i: Idiosyncratic component (obligor-specific risk), assumed to follow a standard normal distribution ($\epsilon_i \sim N(0, 1)$) and independent of $\tilde{F}_i$ and other ϵ_j.

Systematic Factors

Now, let us consider **systematic factors** $\tilde{F}_i$.

$$\tilde{F}_i = \mathbf{a_i}'\mathbf{F} = a_{i1}F_1 + a_{i2}F_2 + \cdots + a_{ip}F_p \tag{9}$$

- $\tilde{F}_i$: This is the systematic risk component for asset i. $\mathrm{var}(\tilde{F}_i) = 1$.

- $\mathbf{a_i}$: This is a vector of factor weight vector for asset i. It determines how much asset i is influenced by each of the common factors in the vector $\mathbf{F}$.

- $\mathbf{F}$: This is a vector of common factors. These factors could represent broad economic variables such as market conditions, industry performance, etc. that affect all assets in the portfolio.

- $F \sim N_p(0, \Omega)$: The common factor vector F follows a multivariate normal distribution with a mean vector of zeros and a covariance matrix Ω.

Default as a Threshold Event

Default rule

$$Y_i = \begin{cases} 1, & \text{if } X_i \leq d_i \\ 0, & \text{otherwise} \end{cases}$$

Link to probability of default

$$P(X_i < d_i) = p_i$$

- X_i : latent credit quality of obligor i
- d_i : default threshold
- p_i : probability of default

Interpretation

- Lower values of X_i correspond to weaker credit quality
- Default occurs when the threshold is crossed
- The threshold d_i is chosen so that

$$P(X_i < d_i) = p_i$$

Asset Correlation

The **asset correlation** $\rho(X_i, X_j)$ can be derived as following:

First, we compute covariance between X_i and X_j. Since the idiosyncratic terms ϵ_i and ϵ_j are independent, they do not contribute to the covariance. And $\mathrm{cov}(\tilde{F}_i, \tilde{F}_j) = a_i' \mathrm{var}(F) a_j = a_i' \Omega a_j$:

$$\begin{aligned} \mathrm{cov}(X_i, X_j) &= \mathrm{cov}(\rho_i \tilde{F}_i + \sqrt{1-\rho_i^2}\epsilon_i,\ \rho_j \tilde{F}_j + \sqrt{1-\rho_j^2}\epsilon_j) \\ &= \rho_i \rho_j \mathrm{cov}(\tilde{F}_i, \tilde{F}_j) = \rho_i \rho_j a_i' \Omega a_j \end{aligned}$$

Thus, the asset correlation is:

$$\rho(X_i, X_j) = \frac{\mathrm{cov}(X_i, X_j)}{\sqrt{\mathrm{var}(X_i)\mathrm{var}(X_j)}}$$

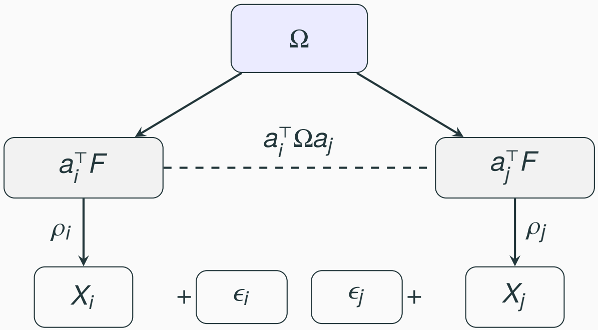
Given that $\mathrm{var}(\tilde{F}_i) = 1$ and therefore, X_i and X_j have unit variance,

$$\rho(X_i, X_j) = \mathrm{cov}(X_i, X_j) = \rho_i \rho_j a_i' \Omega a_j \tag{10}$$

Asset Correlation

Key intuition

Dependence between firms arises share exposure to the same systematic risk factors.



One-Factor Model

If we assume there is only one **systematic factor** F for all the obligors:

$$\tilde{F}_i = F_1 = F_2 = \cdots = F_m = F \tag{11}$$

For obligor i's creditworthiness:

$$X_i = \rho_i F + \sqrt{1 - \rho_i^2}\, \epsilon_i \tag{12}$$

where F is a common factor (e.g., "the economy"). $F \sim N(0, \sigma^2)$.

Real-World Complexities

- **Oil vs Tech**: Both affected by GDP, but differently by oil prices
- **Geographies**: US and German firms react differently to EUR/USD rates

Homogeneous Portfolio

Now, we consider a special case of **one-factor model**, where $\tilde{F}_i = F$ for all i. Moreover, a **homogeneous portfolio**, where the obligors have the same asset correlation, denoted by $\rho_i = \rho$. The critical variable X_i takes the form:

$$X_i = \rho F + \sqrt{1 - \rho^2}\epsilon_i \tag{13}$$

where $F \sim N(0, 1)$ represents the common systematic risk factor, and $\epsilon_i \sim N(0, 1)$ are idiosyncratic risk terms that are independent of F and of each other. Since both F and ϵ_i are standard normal, it follows that $X_i \sim N(0, 1)$.

An obligor i is considered to default if X_i falls below a threshold d_i, which corresponds to their default probability:

$$P(X_i < d_i) = p_i, \quad \text{where } d_i = \Phi^{-1}(p_i).$$

Homogeneous Portfolio

Task 4: Default behaviors of obligors

Simulating the default behavior of a portfolio of 1000 obligors, each with a default probability of 1%. Two scenarios are considered:

- independent defaults
- correlated defaults, where defaults are influenced by a systemic factor with ρ of 0.25

Simulate both scenarios for 2000 times and visualize the result through histograms.

Parameters:

`N = 2000, P(Y_i) = 1%, m = 1000, np.random.seed(42).`

Homogeneous Portfolio

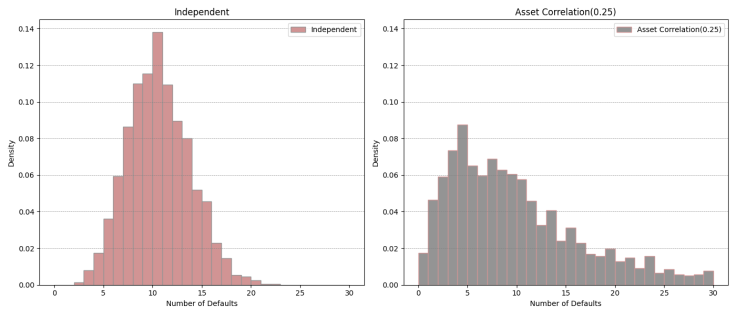


Figure 5: Simulation of the Default Numbers of different dependency structures

Independent defaults: The probabilities will be tightly clustered around the mean (10 defaults). Correlated defaults: The probabilities will have a *fatter tail*, indicating a higher likelihood of extreme losses.

Task 5: Estimate the probability of k defaults*

Under the same context, compute the probabilities for each possible number of defaults and plot the probabilities for both scenarios to compare the distributions.

HINT: The probabilities of the independent defaults can be computed using the binomial probability density function:

$$P(k) = \binom{1000}{k} (0.01)^k (0.99)^{1000-k}.$$

The probabilities of the correlated default can be estimated empirically from the simulation results by counting the frequency of each number of defaults.

Homogeneous Portfolio

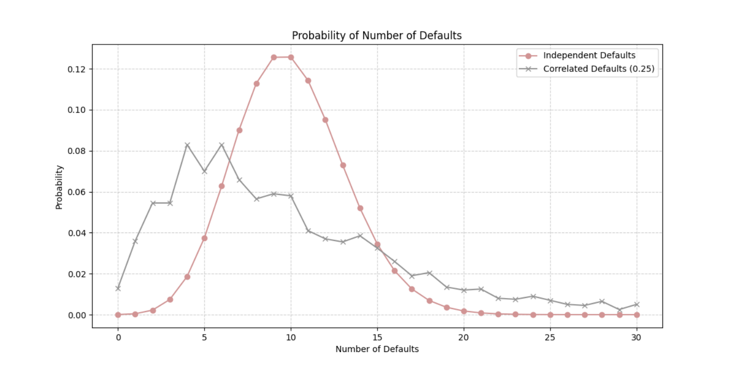


Figure 6: Estimated probability of default numbers

Mixture Model

Factor Models for Default Probabilities (Mixture Models)

Latent variable factor models

$$X_i = \rho_i F + \sqrt{1 - \rho_i^2}\epsilon_i$$

$$Y_i = 1\{X_i \leq d_i\}$$

- Factors drive latent credit quality
- Default occurs when a threshold is crossed

Conditional PD factor models

$$p_i(\Psi) = P(Y_i = 1 \mid \Psi)$$

- Factors directly affect the probability of default
- PD increases when economic conditions deteriorate

Key idea

Factor models can enter either through a latent variable or directly through conditional default probabilities.

Bernoulli Mixture Model

Consider **one obligor** i in the time period $[0, T]$,

the default indicator Y_i is written as $Y_i = \begin{cases} 1 & \text{if default occurs,} \\ 0 & \text{otherwise.} \end{cases}$

The default probability is driven by the values of a factor vector, $\mathbf{\Psi}$, which is a vector representing common risk factors influencing default. The probability of default for obligor i, conditional on $\mathbf{\Psi} = \psi$, is $p_i(\psi)$, where $p_i(\psi)$ is a function mapping the factor vector ψ to the interval $(0, 1)$. Formally:

$$P(Y_i = 1 \mid \mathbf{\Psi} = \psi) = p_i(\psi), \quad P(Y_i = 0 \mid \mathbf{\Psi} = \psi) = 1 - p_i(\psi).$$

The unconditional default probability is given by $p_i = P(Y_i = 1) = \mathbb{E}[Y_i] = \mathbb{E}(p_i(\mathbf{\Psi}))$.

Bernoulli Mixture Model

Consider **a portfolio** of m obligors and fix a time horizon T.

Default Indicator Vector: The random vector $\mathbf{Y} = (Y_1, \ldots, Y_m)'$ is a vector of default indicators for the portfolio.

Factor Vector: $\mathbf{\Psi} = (\Psi_1, \Psi_2, ..., \Psi_p)'$, is a p-dimensional vector of common risk factors that influence the default probabilities of all obligors.

Conditional Independence: The defaults $Y_1, Y_2, ..., Y_m$ are conditionally independent given $\mathbf{\Psi}$, i.e., once $\mathbf{\Psi}$ is known, the default events for different entities are independent of each other.

Bernoulli Mixture Model

Because of the conditional independence, the joint probability of $\mathbf{Y} = \mathbf{y}$, where each $y_i \in \{0, 1\}$, is the product of the individual probabilities:

$$P(\mathbf{Y} = \mathbf{y} \mid \mathbf{\Psi} = \psi) = \prod_{i=1}^{m} p_i(\psi)^{y_i} (1 - p_i(\psi))^{1-y_i}. \tag{14}$$

To get the marginal probability of $\mathbf{Y}$, we integrate over the distribution of $\mathbf{\Psi}$:

$$P(\mathbf{Y} = \mathbf{y}) = \int \prod_{i=1}^{m} p_i(\psi)^{y_i} (1 - p_i(\psi))^{1-y_i} g(\psi)\, d\psi \tag{15}$$

where $g(\psi)$ is the probability density of the factors.

Homogeneous Portfolio

It is often useful to work with a one-factor model. We consider a random variable Ψ and a function $p_i(\Psi)$ such that conditional on Ψ, the default indicator vector $\mathbf{Y}$ is a vector of independent Bernoulli random variables with

$$P(Y_i = 1 \mid \Psi = \psi) = p_i(\psi)$$

In the exchangeable case, where the default probabilities are identical across all obligors in the portfolio given the underlying factor, we have $p_1(\Psi) = p_2(\Psi) = \cdots = p_m(\Psi) = p(\Psi)$. Now, we introduce a new random variable $Q := p(\Psi)$. The distribution function of Q is $G(q)$. Then we have

$$\pi = P(Y_i = 1) = E(Y_i) = E(E(Y_i \mid Q)) = E(Q)$$

$$\pi_k = P(Y_{i_1} = 1, \dots, Y_{i_k} = 1) = E(Q^k) = \int_0^1 q^k dG(q)$$

Homogeneous Portfolio

In an exchangeable model, the default probability and the higher-order joint default probabilities are moments of the distribution of Q. Conditional on $Q = q$, the unconditional distribution of the number of defaults M is:

$$P(M = k) = \binom{m}{k} \int_0^1 q^k (1-q)^{m-k} dG(q). \tag{16}$$

Recall that the default correlation between two firms $i \neq j$ is defined to be the correlation between the default indicators Y_i and Y_j. In exchangeable Bernoulli mixtures, we have:

$$\mathrm{cov}(Y_i, Y_j) = \pi_2 - \pi^2 = \mathrm{var}(Q) \geq 0.$$

The default correlation is given by:

$$\rho_Y := \mathrm{corr}(Y_i, Y_j) = \frac{\pi_2 - \pi^2}{\pi - \pi^2} = \frac{\mathrm{var}(Q)}{E(Q) - E(Q)^2}.$$

Beta Mixing Distribution

Consider that Q follows **Beta distribution**. The density of a beta distribution is given by

$$g(q) = \frac{1}{B(a,b)} q^{a-1}(1-q)^{b-1}, \quad a, b > 0,\ 0 < q < 1,$$

where $B(a,b)$ denotes the beta function. In particular,

$$\pi = \frac{a}{a+b}, \quad \pi_2 = \frac{a+1}{a+b+1} \cdot \pi, \quad \rho_Y = \frac{1}{a+b+1}.$$

The random variable M has a so-called beta-binomial distribution:

$$\begin{aligned} P(M = k) &= \binom{m}{k} \frac{1}{B(a,b)} \int_0^1 q^{k+a-1}(1-q)^{m-k+b-1} dq \\ &= \binom{m}{k} \frac{B(a+k, b+m-k)}{B(a,b)}. \end{aligned}$$

Task 6: Beta-Bernoulli Defaults

Simulate the default behavior of a homogeneous portfolio of 1000 obligors, each with an unconditional default probability of 1%. We consider dependent defaults with default correlation 0.5%. Dependence is introduced through a common random default probability:

$$P \sim \text{Beta}(\alpha, \beta), \qquad Y_i \mid P \sim \text{Bernoulli}(P).$$

Simulate this model 2000 times and visualize the distribution of the number of defaults using a histogram.

Parameters:

`N = 2000, np.random.seed(42).`

Homogeneous Portfolio

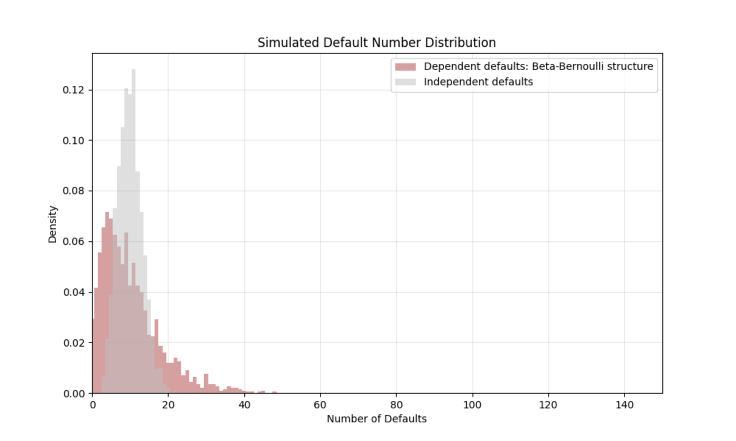


Figure 7: Estimated probability of default numbers

Probit-normal Mixing Distribution

Consider **probit-normal mixing distribution**, where $Q = \Phi(\mu + \sigma\Psi)$ where $\Psi \sim \mathcal{N}(0,1)$, $\mu \in \mathbb{R}$ and $\sigma > 0$. Φ is standard normal distribution function.

Recall that in one factor threshold model, the critical variable X_i takes the form: $X_i = \rho_i F + \sqrt{1-\rho_i^2}\epsilon_i$. If we write the factor $\Psi = -F$, which makes the conditional default probabilities increasing in the factor. Then, we have:

$$
\begin{aligned}
p_i(\psi) &= P(Y_i = 1 \mid \Psi = \psi) = P(X_i \le d_i \mid \Psi = \psi) \\
&= P(X_i \le d_i \mid F = -\psi) = P\left(\sqrt{1-\rho_i^2}\varepsilon_i \le d_i + \rho_i\psi\right) \\
&= \Phi\left(\frac{d_i + \rho_i\psi}{\sqrt{1-\rho_i^2}}\right) = \Phi\left(\frac{\Phi^{-1}(p_i) + \rho_i\psi}{\sqrt{1-\rho_i^2}}\right)
\end{aligned}
$$

Probit-normal Mixing Distribution

Taking the probit transformation gives

$$\Phi^{-1}\big(p_i(\Psi)\big) = \underbrace{\frac{\Phi^{-1}(p_i)}{\sqrt{1-\rho_i^2}}}_{\mu_i} + \underbrace{\frac{\rho_i}{\sqrt{1-\rho_i^2}}}_{\sigma_i}\Psi.$$

Therefore,

$$\Phi^{-1}\big(p_i(\Psi)\big) \sim N(\mu_i, \sigma_i^2), \qquad \mu_i = \frac{\Phi^{-1}(p_i)}{\sqrt{1-\rho_i^2}}, \quad \sigma_i^2 = \frac{\rho_i^2}{1-\rho_i^2}.$$

If we impose further assumption of exchangeable model, we get $\mu = \frac{\Phi^{-1}(p)}{\sqrt{1-\rho^2}}, \quad \sigma^2 = \frac{\rho}{\sqrt{1-\rho^2}}$. And it is equivalent to an exchangeable version of one factor Gaussian threshold model.

Loss Distribution

Approaches to Risk Measurement

Notional-amount approach. Risk of a portfolio = summed notional values of the securities times their riskiness factor.

Comments: The notional-amount approach is simple and conservative, but it overestimates risk for diversified or hedged portfolios and fails to capture true risk exposure.

Loss Distributions. Most modern risk measures are characteristics of the underlying loss distribution over some predetermined time horizon Δt.

Examples: Variance, Value-at-Risk, Expected Shortfall.

Approaches to Risk Measurement

Comments:

- Flexibility and scalability, allow application from individual portfolios to overall institutional risk.
- They account for diversification and netting effects, making them more robust than simplistic notional-based risk measures.
- Their effectiveness depends heavily on accurate loss distribution estimation, which is challenging.
- Estimation often relies on historical data, which may not capture unprecedented events or structural market shifts.
- Model risk and estimation errors can lead to misleading risk assessments.
- These weaknesses are especially critical in extreme market conditions, where tail risks may be underestimated.

Scenario-based risk measures. Scenario-based risk measures are typically considered in stress testing.

Comments: These approaches are particularly valuable for capturing tail risks and black swan events, which may not be evident in past market behavior but can have severe consequences. Additionally, they play a crucial role in regulatory compliance.

However, a key drawback of these methods is their subjectivity, as the accuracy and relevance of results depend heavily on the selection of stress scenarios and underlying assumptions. Furthermore, stress tests do not provide clear probability estimates, making it difficult to quantify the likelihood of extreme losses.

General Setting Review

We consider a portfolio of m obligors and fix a time horizon Δt.

The random vector $\mathbf{Y} = (Y_1, \ldots, Y_m)'$ is a vector of default indicators for the portfolio.

The marginal default probabilities are denoted by $p_i = P(Y_i = 1)$, where $i = 1, \ldots, m$. The default probabilities conditional on critical variable is denoted as $p_i(F_i)$.

The exposure at default for each obligor is denoted as EAD_i.

The weight of each loan is represented by w_i; for simplicity, each obligor holds exactly one loan.

The loss given default for each obligor is LGD_i, also expressed as a percentage.

Task 7: Expected loss and loss distribution

A portfolio consists of 100 obligors, with the first 50 having a probability of default of 0.5%, and the remaining 50 having a probability of default of 0.2%. The default events are independent among obligors. The LGD is 0.4 for the first group of obligors and 0.3 for the second group. The weight of a loan is $\frac{1}{100}$, each obligor has exactly one loan. What is the expected loss during this period? Furthermore, what would the loss distribution look like? Simulate the process 2000 times and generate a plot based on the results.

Example of Loss Distribution

The expected portfolio loss is given by

$$EL = \sum_{i=1}^{m} \mathbb{E}[w_i \cdot LGD_i \cdot Y_i]$$

where Y_i is th default indicator for obligor i. Since Y_i is a Bernoulli random variable with the probability of default p_i, we can write:

$$EL = \sum_{i=1}^{m} w_i \cdot LGD_i \cdot p_i$$

Example of Loss Distribution

Expected Loss:

$$EL = \sum_{i=1}^{m} w_i \times p_i \times LGD_i$$

Breakdown by Groups:

- First 50 obligors:

$$EL_1 = \frac{1}{100} \times 50 \times (0.005 \times 0.4) = 0.5 \times 0.002 = 0.001$$

- Remaining 50 obligors:

$$EL_2 = \frac{1}{100} \times 50 \times (0.002 \times 0.3) = 0.5 \times 0.0006 = 0.0003$$

- Total Expected Loss:

$$EL = EL_1 + EL_2 = 0.001 + 0.0003 = 0.0013$$

Example of Loss Distribution

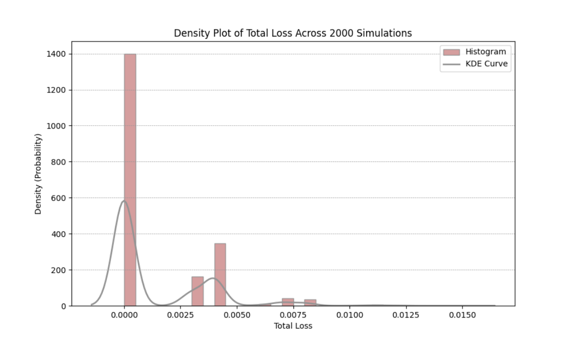


Figure 8: Simulation of Loss Distribution in a Portfolio with no Interdependence: Two Types of Obligors

Mapping of Risks

Consider a portfolio with value of V_t at time t. The change in value of the portfolio during the time period Δt is given by:

$$\Delta V_{t+1} = V_{t+1} - V_t$$

We define the (random) loss as the sign-adjusted value change:

$$L_{t+1} = -\Delta V_{t+1}$$

The distribution of L_{t+1} is called the **loss distribution**.

For longer time intervals, $\Delta V_{t+1} = \frac{V_{t+1}}{(1+r)} - V_t$ (where r is the risk-free interest rate) would be more appropriate, but we will mostly neglect this issue.

Mapping of Risks

In threshold model and mixture model, for a portfolio with m obligors, and Y_i is the default indicator, the loss distribution is

$$L_{t+1} = \sum_{i=1}^{m} w_i \cdot LGD_i \cdot Y_i$$

In the **threshold model**, the default event Y_i for the i-th obligor is determined by comparing a critical variable (e.g., the obligor's asset value or financial health metric) to a threshold (e.g., a default boundary). The randomness in Y_i comes from the uncertainty in the critical variable. For example, if the critical variable is modeled as a normally distributed random variable, the probability of default $P(Y_i = 1)$ can be derived from its distribution.

Mapping of Risks

In the **mixture model**, the default event Y_i is modeled directly as a random variable, conditioned on a market or economic state Ψ (latent variable). Here, the default probability is explicitly expressed as a function of the market condition. The portfolio loss is a random variable because it depends on the random default indicators Y_i, which are driven by the market condition Ψ. The loss distribution is constructed by simulating or analytically deriving the joint distribution of the Y_i's, conditional on Ψ.

Mapping of Risks

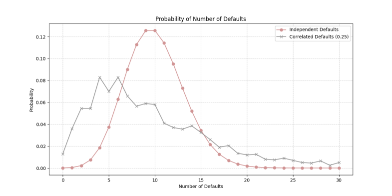


Figure 9: Estimated probability of default numbers

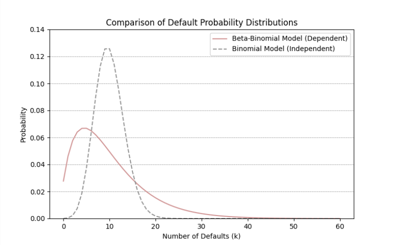


Figure 10: Comparision of Loss Distribution of different dependence structures

QUESTIONS: What are the expected losses for the next period? How would a bank regulate such portfolios to mitigate potential defaults?

Intuitive Explanation: Value at Risk (VaR) answers the question: What is the maximum loss I can expect over a given time horizon with a certain level of confidence?

For example, a 1-day 95% VaR of $1 million means that there is a 95% chance that the portfolio will not lose more than $1 million in one day. Conversely, there is a 5% chance that the loss could exceed $1 million.

Mathematical Definition: Let L be the random variable representing the loss of a portfolio. The VaR at confidence level α (e.g., 95%) is the smallest loss x such that the probability of the loss exceeding x is at most $1 - \alpha$. Mathematically:

$$\mathrm{VaR}_\alpha(L) = \inf\{x \in \mathbb{R} : P(L > x) \leq 1 - \alpha\}$$

Alternatively, VaR can be defined as the α-quantile of the loss distribution:

$$\mathrm{VaR}_\alpha(L) = F_L^{-1}(\alpha) = \inf\{x \in \mathbb{R} : F_L(x) \geq \alpha\} \tag{17}$$

where F_L is the cumulative distribution function of the loss L, and F_L^{-1} is the inverse (quantile) function, if it exists.

Task 8: VaR under the one factor model

Simulate the loss distribution for a portfolio of 100 obligors under a homogeneous one-factor threshold model. In this model, the default probability for each obligor is 3%, the ρ is 0.5, the LGD is 0.4, and the w is 0.01. Perform 2000 simulations of the default process to generate the loss distribution. Try to calculate the VaR analytically under the assumption of a normal distribution.

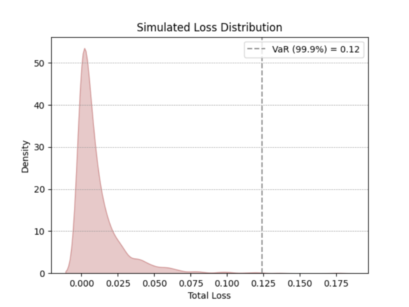


Figure 11: Conditional Expected Loss distribution of simulation data

Simulate the critical variable for each obligor → Assess whether defaults occur→ Calculate the loss in case of default→ Aggregate the losses to determine the total portfolio loss→ Repeat the process 2000 times→ Generate the density plot

Recall that Under the one-factor Gaussian model, the conditional default probability for the i-th obligor, given the systematic factor $F = \bar{F}$, is:

$$p_i(\bar{F}) = \Phi\left(\frac{\Phi^{-1}(p_i) - \rho_i \bar{F}}{\sqrt{1-\rho_i^2}}\right)$$

The portfolio loss $\bar{L}$, conditional on $F = \bar{F}$, is given by:

$$\bar{L} = l(\bar{F}) = \sum_{i=1}^{m} w_i \cdot LGD_i \cdot p_i(\bar{F})$$

The portfolio loss L is a deterministic and monotonically decreasing function of the systematic factor F.

Value at Risk

The monotonic relationship allows us to calculate the quantile of the portfolio loss analytically.

To calculate the quantile of the portfolio loss at confidence level α, we can use the fact that α-quantile of $\bar{L}$ corresponds to the value of $\bar{L}$ when $\bar{F}$ is at its $(1-\alpha)$-quantile.

Mathematically, the α-quantile of L is:

$$\mathrm{VaR}_\alpha = l\left(\Phi^{-1}(1-\alpha)\right)$$

Let's calculate the 99.9% VaR ($\alpha = 0.999$) for the portfolio:

Step 1: Compute $F_{0.999}$

$$F_{0.999} = \Phi^{-1}(1 - 0.999) = \Phi^{-1}(0.001) \approx -3.0902$$

Step 2: Compute $p_i(F_{0.999})$

$$p_i(F_{0.999}) = \Phi\left(\frac{\Phi^{-1}(0.03) - 0.5 \cdot (-3.0902)}{\sqrt{1 - 0.5^2}}\right) \approx 0.349$$

Step 3: Compute $\mathrm{VaR}_{0.999}$

$$\mathrm{VaR}_{0.999} = \sum_{i=1}^{100} 0.01 \cdot 0.4 \cdot 1 \cdot 0.349 = 0.001396$$

Value at Risk

The discrepancy between the simulated VaR and the analytically calculated VaR can arise due to several reasons.

- Idiosyncratic risk not captured in the analytical formula. This additional randomness can lead to a wider range of possible losses, which might result in a smaller VaR compared to the analytical formula.
- Finite number of simulations (2000) leads to underestimation of the tail, especially at very high confidence levels.
- If the portfolio is not perfectly homogeneous, the analytical formula might overestimate the VaR because it does not account for diversification effects.

Expected Shortfall

Intuitive Explanation: Expected Shortfall (ES) answers the question: If things go bad (i.e., losses exceed VaR), what is the average loss I can expect?

For example, a 1-day 95% ES of $1.5 million means that, in the worst 5% of cases (where losses exceed the 95% VaR), the average loss is $1.5 million.

Expected Shortfall

Mathematical Definition: The ES at confidence level α is the average loss in the worst $1 - \alpha$ cases. Mathematically:

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \mathrm{VaR}_u(L)\, du$$

Alternatively, it can be expressed as the conditional expectation of the loss given that the loss exceeds the VaR:

$$\mathrm{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha(L)]$$

where $\mathbb{E}$ denotes the expectation operator, $\mathrm{VaR}_\alpha(L)$ is the VaR at confidence level α.

Expected Shortfall

Task 9: VaR and ES

Suppose the loss distribution of a portfolio is as follows:

Loss (L)	Probability
\$0	90%
\$1 million	5%
\$2 million	3%
\$5 million	2%

Calculate the VaR and ES at the confidence level of 95%.

Expected Shortfall

Step 1: Calculate VaR at 95% Confidence

The 95% VaR is the smallest loss such that the probability of exceeding it is at most 5%. From the table, $P(L > 1) = 5\%$, so:

$$\text{VaR}_{0.95}(L) = 1 \text{ million}$$

Step 2: Calculate ES at 95% Confidence

ES is the average loss in the worst 5% of cases. Here, the worst 5% corresponds to losses of \$2 million and \$5 million. Compute the weighted average of these losses:

$$\text{ES}_{0.95}(L) = \frac{(3\% \times 2) + (2\% \times 5)}{5\%} = \frac{0.06 + 0.10}{0.05} = 3.2 \text{ million}$$

Focus: VaR focuses on the threshold loss (e.g., ”What is the maximum loss with 95% confidence?”). ES focuses on the average loss beyond the threshold (e.g., ”What is the average loss in the worst 5% of cases?”).

Sensitivity to Tail Risk: VaR does not provide information about the severity of losses beyond the threshold. ES captures the tail risk by averaging losses in the worst-case scenarios.

Coherence: VaR is not a coherent risk measure because it fails to satisfy the subadditivity property (i.e., the VaR of a combined portfolio can be greater than the sum of individual VaRs). - ES is a coherent risk measure because it satisfies all four coherence properties (monotonicity, subadditivity, positive homogeneity, and translation invariance).

Then What?

After quantifying the risk, it is used to **manage the risk**.

Interpreting the result and comparing it to risk limits. For example, a bank might set a daily VaR limit of $10 million for its trading portfolio. If the computed VaR exceeds this limit, the portfolio must be adjusted to reduce risk.

Taking action to reduce risk (e.g., diversification, hedging).

Allocating capital to cover potential losses. Use VaR to determine the amount of capital that needs to be set aside to cover potential losses. This is especially important for financial institutions to comply with regulatory requirements (e.g., Basel II).

Monitoring and reporting risk to stakeholders. Continuously improving the risk management framework.

CreditMetrics

Introduction

CreditMetrics is a framework for quantifying credit risk in a portfolio of debt instruments.

Developed by J.P. Morgan in 1997.

It estimates the potential changes in the value of a portfolio due to changes in credit quality.

Key components: p, LGD, EAD, Credit Migration Matrix

Rating Transition Matrix

Credit ratings evolve over time based on transition probabilities.

A transition matrix P is used to model the probability of migration from one rating to another.

$$P = \begin{bmatrix} p_{AA,AA} & p_{AA,A} & \cdots & p_{AA,D} \\ p_{A,AA} & p_{A,A} & \cdots & p_{A,D} \\ \vdots & \vdots & \ddots & \vdots \\ p_{D,AA} & p_{D,A} & \cdots & p_{D,D} \end{bmatrix}$$

Rating Transition Matrix

	AAA	AA+	AA	AA-	A+	A	A-	BBB+	BBB	BBB-	BB+	BB	BB-	B+	B	B-	Cs	D
AAA	96.79	2.71	0.42	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00
AA+	6.45	85.16	6.61	1.77	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
AA	0.00	6.22	85.17	6.74	0.52	0.42	0.10	0.52	0.00	0.00	0.00	0.31	0.00	0.00	0.00	0.00	0.00	0.00
AA-	0.00	0.00	7.82	83.45	7.16	0.17	0.50	0.17	0.00	0.17	0.44	0.00	0.00	0.11	0.00	0.00	0.00	0.00
A+	0.00	0.00	0.07	13.35	73.28	9.30	2.03	1.12	0.14	0.63	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.01
A	0.00	0.00	0.00	1.15	12.33	77.29	5.71	1.68	0.77	0.96	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.01
A-	0.00	0.00	0.00	0.00	0.94	11.47	77.82	6.94	0.41	1.57	0.67	0.16	0.00	0.00	0.00	0.00	0.00	0.02
BBB+	0.00	0.00	0.00	0.00	0.00	2.16	12.39	70.86	11.24	2.41	0.60	0.24	0.06	0.00	0.00	0.00	0.00	0.04
BBB	0.00	0.00	0.00	0.00	0.00	0.00	1.87	16.60	68.05	11.16	0.99	0.11	0.00	0.50	0.22	0.11	0.33	0.06
BBB-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.93	14.94	74.69	6.50	2.13	0.27	0.08	0.15	0.12	0.08	0.11
BB+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.54	20.57	66.38	9.93	1.14	0.18	0.06	0.00	1.02	0.18
BB	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.78	14.20	70.80	11.15	1.80	0.68	0.15	0.05	0.40
BB-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.03	10.55	73.79	11.47	1.19	0.46	0.61	0.90
B+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.03	0.92	10.20	68.70	15.30	2.50	0.85	1.46
B	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.61	13.50	70.77	9.84	2.91	2.38
B-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.43	15.07	66.85	8.05	7.59
Cs	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.74	1.81	13.90	32.08	51.47
D	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 1: Normalized Average One-year Sovereign Foreign Currency Transition Matrix from S&P (transition probabilities in %).

Rating Transition Matrix

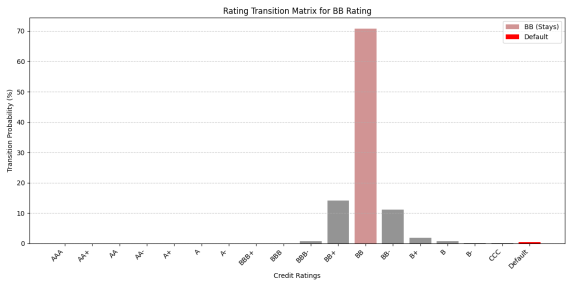


Figure 12: Visualization of Rating Transition for BB Rated obligor

Variance as Credit Risk Measurement

Task 10: Calculate Standard Deviation

The following table shows the year-end ratings, their probabilities, and the corresponding bond values in a portfolio:

Year-end Rating	Probability of State (%)	New Bond Value + Coupon ($)
AAA	0.02	109.37
AA	0.33	109.19
A	5.95	108.66
BBB	86.93	107.55
BB	5.30	102.02
B	1.17	98.10
CCC	0.12	83.64
Default	0.18	51.13

Table 2: Year-end ratings and bond values.

Calculate the standard deviation of the bond values as a credit risk measurement.

Variance as Credit Risk Measurement

For each rating, we get the **Probability-Weighted Value**.

Year-end Rating	Probability of State (%)	New Bond Value + Coupon ($)	Probability Weighted Value ($)
AAA	0.02	109.37	$0.0002 \times 109.37 = 0.02$
AA	0.33	109.19	$0.0033 \times 109.19 = 0.36$
A	5.95	108.66	$0.0595 \times 108.66 = 6.47$
BBB	86.93	107.55	$0.8693 \times 107.55 = 93.49$
BB	5.30	102.02	$0.0530 \times 102.02 = 5.41$
B	1.17	98.10	$0.0117 \times 98.10 = 1.15$
CCC	0.12	83.64	$0.0012 \times 83.64 = 0.10$
Default	0.18	51.13	$0.0018 \times 51.13 = 0.09$

Table 3: Probability-weighted values.

The **mean value** is the sum of all **Probability Weighted Values**. The **standard deviation** is then the square root of the variance, which is the average of the squared deviations from the mean.

$$\text{Standard Deviation} = \sqrt{\sum(\text{Probability Weighted Difference Squared})}$$
$$= \sqrt{\text{Variance}} = \sqrt{8.9477} = 2.99$$

Quantile as Credit Risk Measurement

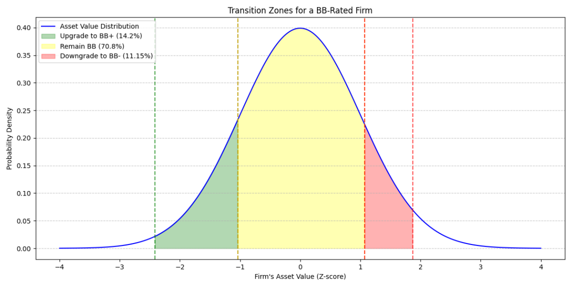


Figure 13: Credit Rating Thresholds for BB-Rated Obligor

The rating setting in CreditMetrics is closely linked to the threshold model through the concept of default probabilities and rating transition probabilities.

Each credit rating corresponds to a specific probability of default over a given time horizon, and these probabilities are used to define the thresholds in the threshold model.

For example, if a B rated obligor has a 1% probability of default over the next year, this PD can be mapped to a specific threshold on the critical variable y_i using the inverse of the standard normal cumulative distribution function. Specifically, if $\Phi^{-1}(0.01) = -2.33$, then the default threshold for the B rated obligor is set at $C = -2.33$.

Mathematically, the critical variable X_i is often modeled as:

$$X_i = \beta_{\mathbf{i}}\mathbf{F} + \eta_i \epsilon_i \tag{18}$$

where $\mathbf{F}$ is a vector of systematic risk factors (e.g., GDP growth, interest rates). $\beta_{\mathbf{i}}$ represents the sensitivity of $\mathbf{F}$ to the systematic factors. ϵ_i is the idiosyncratic risk factor, typically assumed to follow a standard normal distribution.

The probability of being in a particular rating bin is determined by the cumulative distribution function of X_i:

$$P(C_k \leq X_i < C_{k+1}) = \Phi(C_{k+1}) - \Phi(C_k)$$

If we restrict the set of outcomes to two states, default and non-default, (18) can be represented as an equivalent threshold model.

CreditMetrics accounts for rating transitions, defaults, and correlations in a portfolio.

The model can compute the potential loss in portfolio value due to credit risk over a specified risk horizon.

The framework allows for stress testing by adjusting the systematic risk factors to simulate adverse economic scenarios.